# S Æ T E R N E S D Æ G v1.2.0

created by k.odk

01 · THE IDEA

# Made to be played

Sæternesdæg is made to be played, a little like a modular synthesizer patch that lives in your hand. You shape a single voice by touch, by light and by the angle you hold the phone at, and listen to where it goes. It is happiest treated as one instrument among your others, rather than as a place to arrange a whole song.

At its heart is a chain of four modules: an oscillator, a folder, a pulse divider and an amplitude XOR. The end of the chain is sent back to the start. From that moment the voice stops being a waveform and becomes a small system with a will of its own — it settles, it breaks, it finds new places. Your part is to lean on it and catch what you like.

Almost everything that moves inside the instrument can be sent somewhere else. Much of this guide is about that: **who sends, who receives**, and how to read a patch at a glance.

**Too much to learn? Skip it.** Hold the random keys in the SETUP panel and simply enjoy how it behaves — the way the loop settles, breaks apart and finds somewhere new is the whole point of this instrument. Before you hold them, look at the **PROBABILITY** meter above the keys: it decides where the random pitches land. Keep its hills to the left, in the low, slow range — that is where Sæternesdæg sounds best. Toward the right, the fast range tends to turn into noise.

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# What it needs

Sæternesdæg runs four modules, three oscillators, a hyperchaos oscillator, filters and effects in real time, oversampled, while it watches the camera and reads the motion sensors. It asks more of the device than a typical app.

## System

| | |
|---|---|
| IOS | iOS 18.0 or later. |
| DEVICE | iPhone and iPad, played in landscape. On iPad the whole instrument fills the screen and every panel opens full size. |
| RECOMMENDED | iPhone 13 or newer. Sæternesdæg is developed and tested on an iPhone 13. Older iPhones that run iOS 18 will open it, but may need lower oversampling, or the camera off, to play without crackles. |
| SENSORS USED | Camera (front or back) and the motion sensors (tilt and the gravity ball). Both are optional — the instrument plays without them — and iOS asks your permission the first time. |
| LISTENING | Headphones or good speakers. The loop produces very low and very high content that phone speakers cannot reproduce. |
| PRIVACY | Nothing leaves the device. The camera picture is analysed for brightness and movement and is never stored or sent; the app does not use the network. |

## Performance and heat

The ENGINE row in the SETTINGS panel shows what the phone is doing:

- **Load dot** — how much of the audio deadline the sound engine uses, green, yellow or red, with the figure beside it. Over 100% the sound breaks up.
- **Oversampling** (1 to 4) — higher is cleaner and heavier. Lower it first if you hear crackles.
- **Temperature**, and **SAFE** — SAFE caps the oversampling, turns off the pad edge glow and holds the camera at 15 fps.
- Lowering the camera frame rate also cuts the heat, and makes the movement step.

## About AUv3

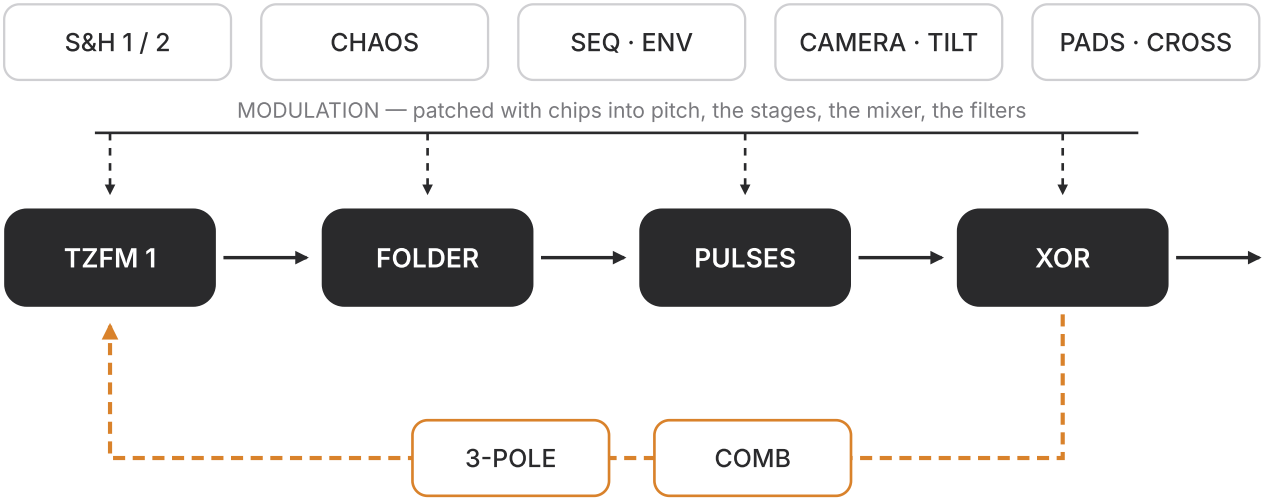
Sæternesdæg is a **standalone app**, not an AUv3 plug-in. The camera, the tilt of the phone and the gravity ball are part of the instrument itself, and a plug-in inside a host app cannot use them in the same way. It keeps playing in the background.

That said, we know many of you want to use it in finished pieces — to add improvised colour and texture to a track. With that in mind, **an AUv3 version is under consideration**. News will come through the newsletter: buttondown.com/kodk

03 · THE SIGNAL PATH

# Four modules in a row

One oscillator, four shaping stages in series, after circuits by Ian Fritz. What comes out of the last stage returns to the oscillator — and the series becomes a living thing.




The return path — AC coupled, so it never settles for good. 3-POLE and COMB can be placed inside it.

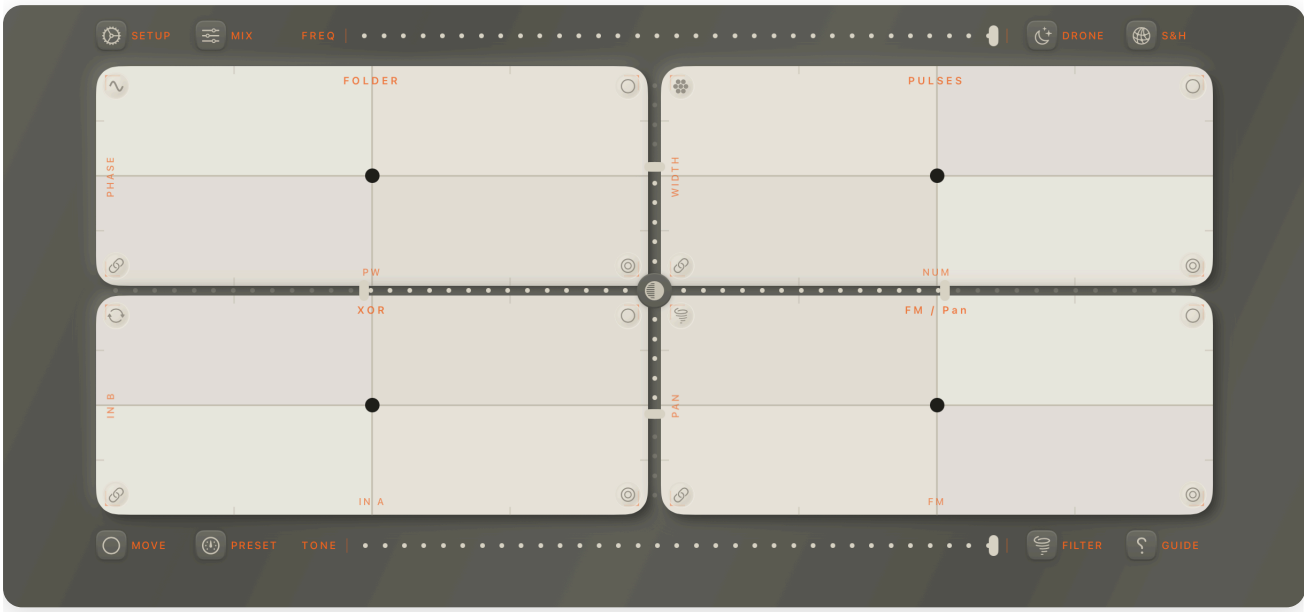
THE SOUND YOU HEAR

MIXER · TZ 1–3 → REVERB · CHORUS → EQ · COMP → LIMITER → TONE → OUT

TZFM 2 and TZFM 3 (the SUB panel) join at the mixer. They never enter the loop.

| | |
|---|---|
| **TZFM** | A through-zero FM oscillator. With deep linear FM the phase increment goes negative and the phase genuinely runs backwards. |
| **FOLDER** | Turns the part of the input above a threshold into an antisymmetric pulse, and fills its inside with a wavefolder. |
| **PULSES** | Slices the voltage with a chain of comparators — one to five pulses per cycle. |
| **XOR** | Judges the *sum* of the PULSES output and the returning signal with a window — by amplitude, not polarity. The heart of the circuit. |

# Where your hands go




### The keys

| | |
|---|---|
| **SETUP** | SETTINGS — camera, ball, engine, presets, random keys |
| **MIX** | MIXER — TZFM 1 inputs, mixer strips, reverb, chorus |
| **DRONE** | SUB — TZFM 2 and TZFM 3 |
| **S&H** | MOD — two S&H, CHAOS, CROSS |
| **MOVE** | The recorder for all four pads |
| **PRESET** | BUILD — output stage and sequencer |
| **FILTER** | FILTER + ENV — 3-POLE, COMB, envelope follower |
| **GUIDE** | The manual inside the app |

### The two long sliders

**FREQ** (top) is the base frequency of TZFM 1. Everything else moves around it. **TONE** (bottom) is a one-pole low-pass on the output, 200 Hz to 18 kHz — the last place to take the edge off.

### The cross of sliders

Horizontal left: the **XOR threshold**, sweeping the window past both ends of the swing. Horizontal right: how much **XOR** is in the output. Vertical bottom: how much **PULSES** is mixed in. Vertical top: the **trigger threshold** — centre is high and rarely hit, outward is low and hit often. This one knob decides how the thing goes wrong.

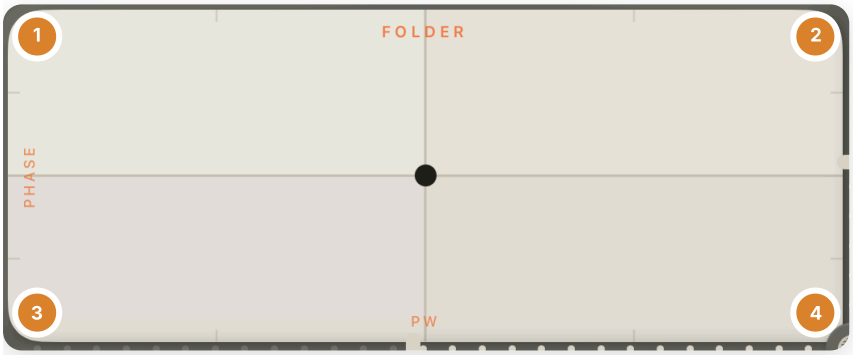
All four can be modulated hands-free — see CROSS, section 09.

# Four squares, eight hands

| PAD | X / Y | SENDS AS |
|---|---|---|
| 1 · FOLDER | Pulse width / phase — phase offsets the threshold asymmetrically and breaks the spacing of the two pulses | P1 |
| 2 · PULSES | Number of pulses (1 to 5, continuous) / pulse width | P2 |
| 3 · XOR | Level of PULSES into the sum / level of the returning signal into the sum. Since the window judges by amplitude, these two levels decide where it sits. | P3 |
| 4 · FM / PAN | How strongly the return acts on the oscillator / how far the pan swings. With PURE on, X reads VOL. | P4 |

Each pad chip sends X; tap it again for Y (same word, dark yellow).

## The buttons on each pad




1. The switch inside it
2. Pad recorder
3. Patch on / off
4. Gravity ball

### The switch inside it

**Pad 1:** which TZFM output feeds the chain — sine, triangle, saw, square; the little glyph is the waveform. **Pad 2:** the PULSES odd-harmonics switch — filled is bipolar, hollow is unipolar and keeps the even harmonics. **Pad 3:** what the XOR judges against — the returning signal (inside the loop, the default) or the FOLDER output (outside it). Pad 4 has no switch.

### Patch on / off

Takes that stage out of the series. Pad 1 FOLDER, pad 2 PULSES, pad 3 XOR, pad 4 the return cable itself. Orange is connected, grey unplugged. All four off and only the raw TZFM is left — with the sine selected, a plain sine.

### Recorder

Records that pad's finger movement and loops it, keeping the timing — speed and pauses survive. Tap to step idle → recording → playing → idle; double-tap erases. **MOVE**, in the bottom row, records all eight values together at 30 per second — camera, ball and fingers alike — and gives up if nothing moved.

### Ball

Tilt becomes gravity and the pointer rolls, bouncing off the edges. A finger stops the physics; let go and it flies off with that speed. Gravity, bounce, drag and **JITTER** (a small random shove each frame — a slow wander that never repeats) are in SETTINGS.

### Camera

The picture is cut into 8 × 16 cells. Each pad pulls its pointer toward the cell with the most movement, the brightest or the darkest. **SENS** is the floor under motion. Which pads follow is set in SETTINGS.

# Who sends, who receives

Every patch is made the same way: a row of small chips, one per source or destination, and you tap the one you want. No menus and no cables — one glance across a panel tells you what is plugged into what.

### Tapping a lit chip

It does not go dead. It steps through the rest of its group, and the word on the chip changes to say where you are.

### Reading the colour

orange is the plain side of a group. dark yellow is the flipped side — an inverted output, a Y axis, a second unit.

T1 → T2 → T3

WV → PL → AX → FB

X → X inverted · then Y, Z, U

P1 → P1 its Y — off

## The senders

| CHIP | WHAT IT SENDS |
|---|---|
| T1 T2 T3 | The raw oscillators before any shaping — TZFM 1, then the two in SUB. Audio rate: this is where cross modulation comes from. |
| WV PL AX FB | The four taps along the chain: FOLDER's output, PULSES, XOR, and the returning signal itself. |
| X Y Z U | CHAOS's four outputs; tap again to invert. U is the fastest. |
| SH | An S&H hold — the voltage grabbed at the last trigger. Orange is unit 1, dark yellow unit 2. |
| SQ | The sequencer's staircase, whatever step it is on. |
| EF | The envelope follower. In CYCLE it swings both ways and can drive an oscillator. |
| P1 … P4 | The pads' X, centre being zero; tap again for Y. |
| 1G 1T 2G 2T | The S&H gates and triggers. Only in the envelope follower's IN. |
| — | Nothing patched. The row is off. |

Not every row carries every sender: audio-rate rows leave out the slow ones, and CROSS is cut down to the six worth holding on to.

# Where things can go

## Receivers that pick a sender

| RECEIVER | WHERE |
|---|---|
| Every **cv** row — pitch, VCA cv, mixer knobs, filter, envelope RISE / FALL… | MIXER, SUB, FILTER + ENV |
| The oscillators' audio-rate rows — **LIN FM**, **EXP FM** and the reset input | MIXER (TZFM 1), SUB (TZFM 2 / 3) |
| S&H **TIMING** and **IN** (CHAOS is left out of TIMING) | MOD |
| **CROSS** — the four halves of the cross sliders (X, Y, Z, U, TG, SQ only) | MOD |
| Envelope follower **IN** — an audio node, or 1G 1T 2G 2T | FILTER + ENV |
| Each mixer strip's six knobs — a chip under every knob | MIXER |

## Senders that pick a destination

S&H's three outputs (gate, trigger, hold) and CHAOS's four each pick one destination, at full depth:

| | |
|---|---|
| FREQ | The oscillator's pitch, two octaves either way. |
| PW | FOLDER's pulse width — the same thing pad 1 moves. |
| NUM | How many pulses PULSES cuts per cycle. |
| XOR | The XOR threshold — where the window sits. |
| FM | How strongly the return acts on the oscillator. |
| PAN | How far the sound swings across the stereo field. |

### The sequencer's OUT

One of the four pitches — **T1**, **T2**, **T3**, or **FLT** (the 3-POLE's cutoff) — or one of CHAOS's three inputs, **RTE**, **GAN**, **DMP**.

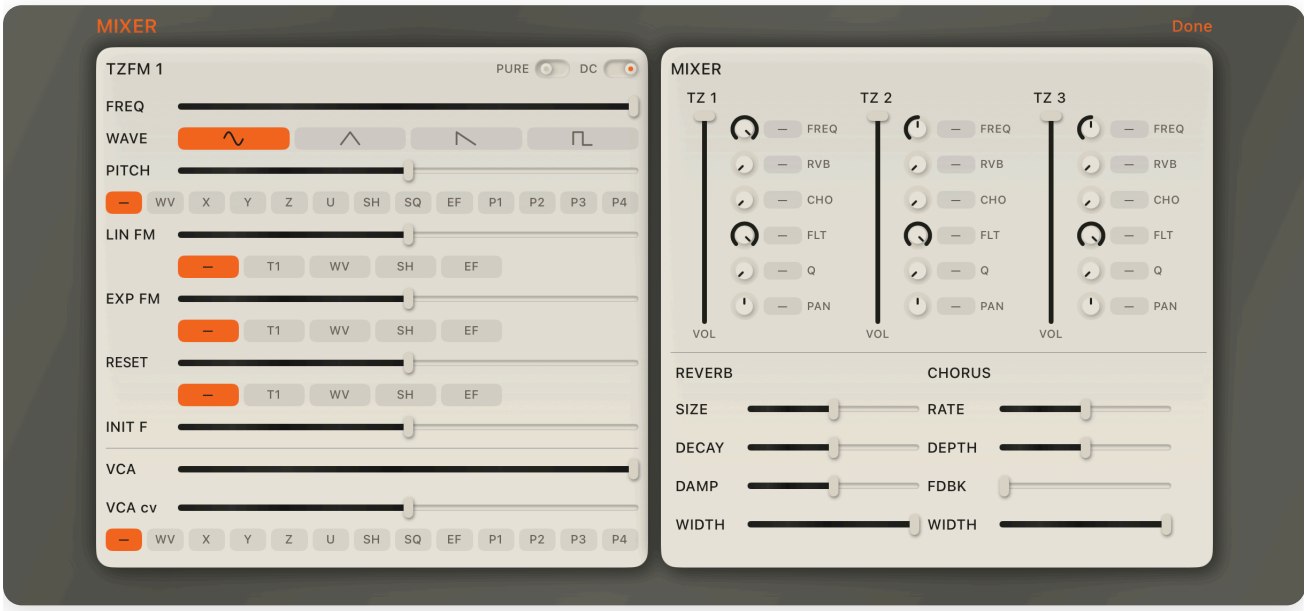
### The 3-POLE's place in the row

**IN** picks where it listens — OSC, WAV, PLS or XOR. **OUT** picks where it returns — **SER** back into the row, **FB** into the return path, **OUT** straight to the mixer, **XOR** into the XOR's sum.

**IN2** gives the filter a second route with its own input, output, mode and level.

07 · MIX — TZFM 1 AND THE MIXER

# Three oscillators, one voice MIX




### TZFM 1

The main oscillator: **FREQ**, **WAVE**, then its sockets — **PITCH**, **LIN FM**, **EXP FM**, the reset input, **INIT F**, the **VCA** and its cv. The audio-rate rows take T1 WV SH EF, so another oscillator gives cross modulation. **DC** is the coupling of the linear FM input; INIT F is the manual offset into that same point.

The VCA does not stop at zero: a cv swinging negative flips the sign of the whole voice — an audio-rate node in VCA cv is ring modulation, not tremolo.

**PURE** parks every stage where it does nothing, so a plain sine comes out; raise the stages back one at a time. Off puts every knob back.

### Mixer

**TZ 1**, **TZ 2**, **TZ 3** — one strip each: **VOL**, then **FREQ**, **RVB** and **CHO** sends, **FLT** and **Q** (a low-pass on the channel) and **PAN**. Each knob has its own chip.

### Reverb and chorus

No switches — they run whenever a strip sends to them. REVERB: **SIZE**, **DECAY**, **DAMP**, **WIDTH**. CHORUS: **RATE** (0.05–6 Hz), **DEPTH**, **FDBK** (past halfway it becomes a flanger), **WIDTH**.

# Two more, outside the loop DRONE



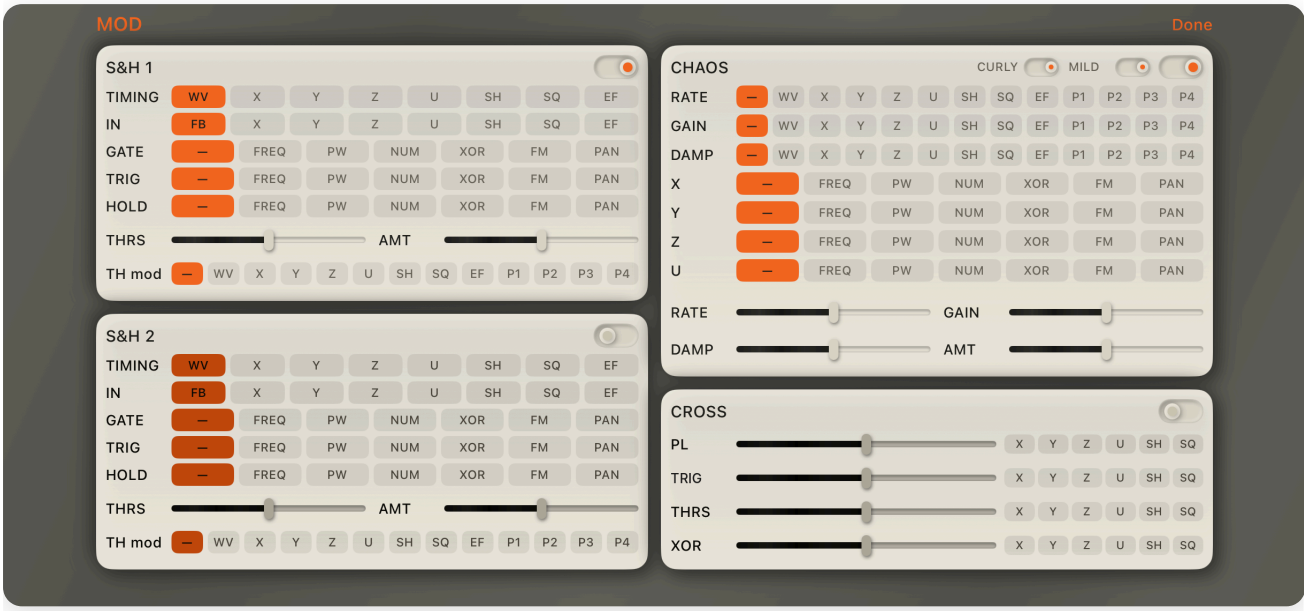

Plain oscillators with no shaper and no return path. Each has **FREQ**, **WAVE**, **PITCH** (cv, two octaves either way), **LIN FM**, **EXP FM**, the reset input, **INIT F**, and a **VCA** with its own cv. The switch on the right turns each one on.

**LIN FM** is through-zero: deep enough and the phase runs backwards. **EXP FM** multiplies the pitch, so harmonics pile up but the note stays.

**DC** decides whether the linear FM input passes its DC. **INIT F**: centre is nothing, right is positive, left negative — the saw falls instead of rising. The reset input has no depth: patched or not.

Every oscillator output is a shared node, so they patch into each other — TZFM 2 into TZFM 3's FM, TZFM 3 into TZFM 2's pitch — and into TZFM 1. They join at the mixer and never enter the loop.

# Where movement comes from S&H




### S&H 1 and 2

Each samples **IN** the moment **TIMING** crosses **THRS** going up, and sends **GATE**, **TRIG** and **HOLD** each to a destination — FREQ, PW, NUM, XOR, FM, PAN. THRS at centre is plain zero-crossing; move it and only the taller part counts. **TH mod** moves the threshold, so hits never settle into a pattern. CHAOS is left out of TIMING on purpose.

### CHAOS

Ian Fritz's fourth-order hyperchaos oscillator. **RATE**, **GAIN**, **DAMP** — each with a source row — and four outputs **X Y Z U**, each to a destination. **CURLY / SQUIRRELLY** is the shape of the attractor, **MILD / WILD** how hard it is driven.

### CROSS

The four halves of the cross sliders, hands-free: **PL** (PULSES mix), **TRIG** (trigger threshold), **THRS** (XOR threshold), **XOR** (XOR mix). The slider is the floor; the source moves it from there.

# Inside the loop, and beside it FILTER




### 3-POLE

Ian Fritz's third-order VCF. **MODE** LLL / LHL / HLH / HHH picks the tap — and the resonance returns from that tap, so the mode moves the self-oscillation as well as the band. **FREQ**, **RES** and their cv rows. **IN**: OSC, WAV, PLS or XOR. **OUT**: SER back into the row, FB into the return path, OUT to the mixer, XOR into the sum. In FB it gets wildest.

**IN2** sends a second signal through its own copy — own IN, OUT, TYPE and level — so one module sits in two places at once.

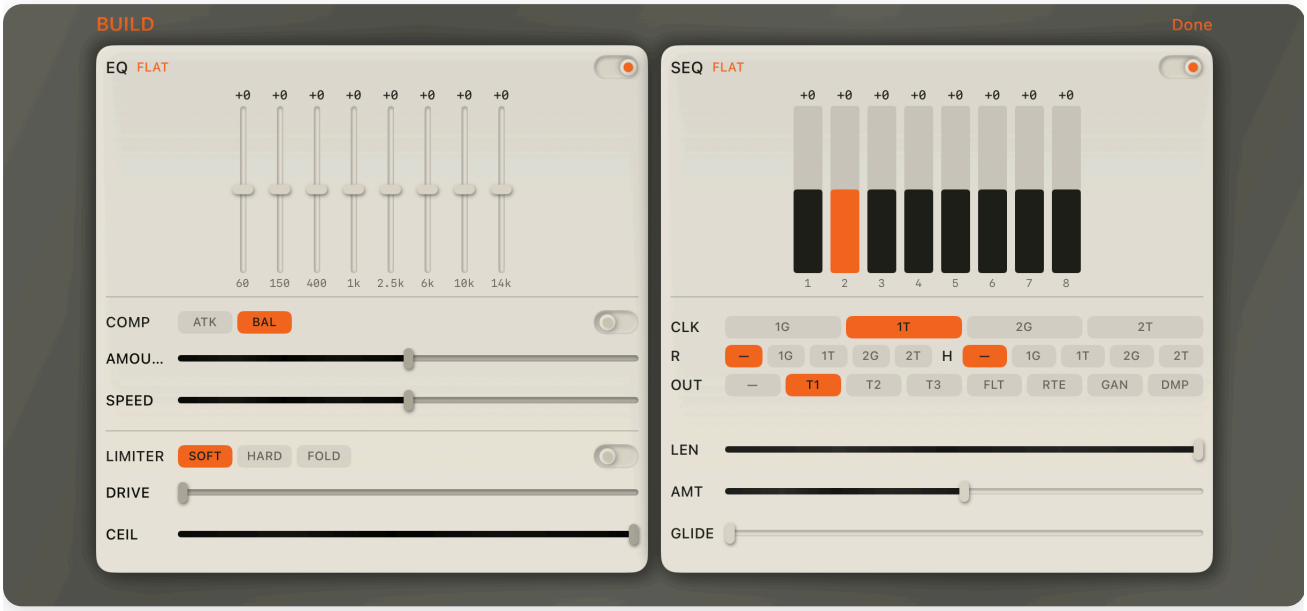
### ENV

Watches whatever is patched to **IN** and sends its shape as EF. Fed a gate (1G 1T 2G 2T), it is an envelope generator. **RISE** / **FALL** 1 ms–30 s, each with cv. **CYCLE** makes it an LFO with two halves you set separately.

### COMB

A tuned delay in the return path — plucked, metallic tones. **PITCH** (with **P cv**), **REGEN**, **DAMP**, **LOOP** (how much stays in the loop) and **OUT**.

# Build the finish, and a staircase PRESET




### Output stage

**EQ** — eight bands with their own switch; **FLAT** levels them.

**COMP** — ATK lets the attack through, BAL evens it out; **AMOUNT** and **SPEED**.

**LIMITER** — SOFT, HARD or FOLD, with **DRIVE** into it and **CEIL** for the ceiling.

None of the random keys touch the output stage or the mixer's VOL.

### SEQ

Eight steps of stepped voltage — no sound of its own. Drag a bar for its height; **FLAT** levels them. **CLK** picks the S&H that clocks it (gate or trigger). **R** resets to step one, **H** holds; reset works while held. **OUT**: T1, T2, T3, FLT (the 3-POLE's cutoff), or CHAOS's RTE, GAN, DMP. **LEN** sets where it wraps, **AMT** its range, **GLIDE** bends the corners. It appears as SQ in every cv list.

12 · SETUP — SETTINGS AND PRESETS

# The machine, and your places SETUP




**ENGINE** — **OS** oversampling (1–4×), temperature letter, load dot and figure; **SAFE** caps the oversampling, drops the pad glow and holds the camera at 15 fps.

**CAMERA** — on/off, **MODE** (motion, bright, dark), **FRONT**, **FPS**, and **SENS**, the floor under motion.

**BALL** — **GRAV**, **BOUNCE**, **DRAG**, and **JITTER**, a small random shove each frame so the ball keeps wandering.

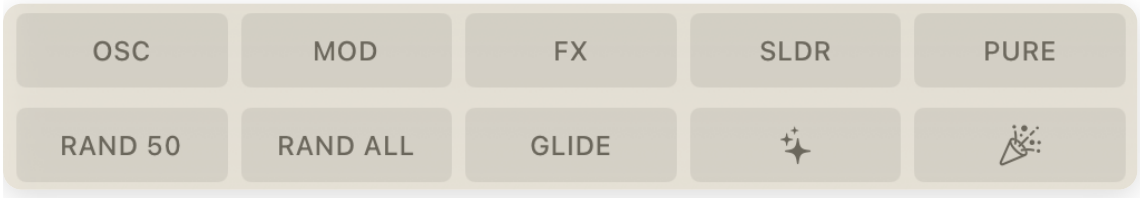
**PRESET** — sixteen slots. **Tap** to recall, **hold** to save; the bin then a slot clears it. A preset keeps everything, including whether the camera and ball were running. **INIT** (hold) puts every panel back.

**PROBABILITY** — five hills decide where random pitches land.

The random keys are on the next page.

13 · RANDOM KEYS

# Losing your place, on purpose



| | |
|---|---|
| **OSC** | the oscillators and mixer strips |
| **MOD** | the modulation and the patching |
| **FX** | the filter and the effects, sends included |
| **SLDR** | the pads and the front sliders |
| **PURE** | the switch from MIX, as a key — the random keys leave it as they find it |
| **RAND 50** | redraws about half the patch, so the sound keeps some of its shape |
| **RAND ALL** | redraws the whole patch |
| **GLIDE** | a switch: knobs drift one at a time; now and then a cable is repatched |
| ✦ | one push — an eighth of RAND ALL, pitch left alone |

### Hold to glide

Cables, switches and waveforms jump at once; everything continuous travels there over about five seconds. Let go early and it arrives at once — a tap is instant, a long press a slow drift.

### The party popper — a gesture

The breath between one sound and the next: a tail into the reverb, a burst of chopping, a roll that speeds into a pitch, a held silence. Its name shows on the key. Tap and it plays its own length; hold and it keeps going. Nothing is left changed when it ends, and the reverb and chorus tails ring on.

14 · CREDITS

# Whose circuits these are

Every stage in this instrument follows a circuit designed by Ian Fritz and published on his own site: the through-zero FM oscillator, the folder, the pulse divider, the amplitude XOR, the third-order filter and the fourth-order chaos oscillator. They are used here with his permission.

*With warm thanks to Ian Fritz, for kindly letting his designs live on in this instrument.*

Sæternesdæg is an independent piece of software, not affiliated with, sponsored by, or endorsed by him. No schematic, board layout, diagram, photograph or text from his site is reproduced in this app or in this guide. If you like what these circuits do, build the real ones. That is where they come from.

## What is mine

The code, the sound engine, the panels, and every word of this manual. The models were written from published descriptions of how the circuits behave and from listening to the result. Nothing here is a port, a netlist, or a copy of a board. Written in Swift with AVAudioEngine, with no third-party audio code.

Sæternesdæg, Old English for Saturday. © k.odk. Sound you make with it is yours.

# Version 1.2.0

CURRENT

## 1.2.0

iPhone and iPad, landscape · iOS 18.0 or later. This guide describes this version.

### What’s new in 1.2.0

| | |
|---|---|
| **iPad** | Sæternesdæg runs on iPad, in landscape and full screen. The whole instrument fills the display, so pads are taller and every key and label is larger. |
| **Panels on iPad** | Every panel opens full screen, keeping exactly the same layout as on iPhone, scaled up to fill the display. |

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